

Dr.-Ing. Max Domagk

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Research Interests

Electrical power systems, power quality, data analytics

Academic & Professional Career

Research Associate, *Technische Universität Dresden* — since Jan. 2010

Chair of Networked Energy Systems (Prof. Martin Wolter; formerly Chair of Electrical Power Supply under Prof. Peter Schegner), Institute of Electrical Power Systems and High Voltage Engineering

Ph.D. (Dr.-Ing.), *Technische Universität Dresden* — Jan. 2010 – Oct. 2015

Degree: Dr.-Ing. (magna cum laude), “Identification and Quantification of Correlative Relationships between the Electrical and Climatic Environment and Power Quality” (urn:nbn:de:bsz:14-qucosa-211866)

Research Internship, *Helsinki University of Technology, Finland* — Apr. 2008 – Sep. 2008

Simulation-based study on decentralized compensation of capacitive earth-fault currents, Department of Electrical Power Systems (Prof. Matti Lehtonen)

Diploma studies (M.Sc.-equivalent), *Technische Universität Dresden* — Oct. 2004 – Dec. 2009

Degree: Dipl.-Ing. in Electrical Engineering, thesis on resynchronization of fault recordings and parameter identification of asymmetric overhead lines on the Dynamic Network Model

Research Projects (selected)

EURAMET (Grant No. 24DIT05), *Project Contributor* — 2025–2028

GridData – Metrology for reliable power grid data analytics

DFG (German Research Foundation) (Project No. 521923789), *Principal Investigator* — 2024–2026

New methods for multivariate analysis of voltage and current quality in large datasets

SysQual (industry-funded), *Project Contributor* — 2023–2026

New methods for efficient and unified monitoring and assessment of power quality in the transmission system (50Hertz, Amprion, TenneT, TransnetBW)

TransnetBW (industry-funded), *Project Lead* — 2022–2023

Systematic study on the realistic simulation of harmonic zero-sequence systems in the transmission network

BMW (Federal Ministry for Economic Affairs) (Grant No. 0325757A-L), *Project Contributor* — 2015–2018

NetzHarmonie: Optimized efficiency and grid compatibility for the integration of generation units from a harmonics perspective

Teaching

Regenerative Energies and Energy Storage, *Guest lecturer, UMIT Tirol, Austria (online)* — since summer term 2022

Artificial Intelligence applied to Smart Grids, *Course developer and instructor* — since winter term 2020/2021

Research Seminar Electrical Power Supply, *Planning and instruction* — winter term 2020/2021

Topic: Image/pattern-recognition methods for classifying power quality time series

Storage of Electrical Energy, *Course developer* — winter term 2015/2016

Duties at the chair (since 2010): supervision of 14 student theses (5 Bachelor's, 9 Diploma/Master's); substitute lecturer for undergraduate courses (incl. Operation of Electrical Power Systems); instructor for laboratory courses (incl. line parameters, switching overvoltages)

Publications (selected)

1. M. Domagk, P. Feistel, J. Meyer, M. Lindner, J. Kilter, "Ensemble Forecasting of Power Quality Parameters," *22nd International Conference on Harmonics and Quality of Power (ICHQP)*, Dresden, Germany, Sep. 2026 (accepted).
2. M. Domagk, J. Meyer, M. Lindner, "Identification and Visualization of Correlation Structures in Large-Scale Power Quality Data," *22nd ICHQP*, Dresden, Germany, Sep. 2026 (accepted).
3. A. Bracale, P. Caramia, P. De Falco, M. Domagk, J. Meyer, "Integration of Clustering Techniques in Probabilistic Current and Voltage Harmonic Forecasting," *IEEE Transactions on Power Delivery*, 2025.
4. O. Zyabkina, M. Domagk, J. Meyer, P. Schegner, M. Lindner, H. Mayer, C. Butterer, "Detection and Characterisation of Atypical Harmonic Patterns in Big Power Quality Data," *IET Generation, Transmission & Distribution*, vol. 19, no. 1, e70062, Apr. 2025.
5. M. Domagk, O. Zyabkina, J. Meyer, M. Lindner, "Uncertainty in the Assessment of Harmonic Voltages and Currents in Sequence Domain," *NEIS 2025 – Conference on Sustainable Energy Supply and Energy Storage Systems*, Hamburg, Germany, Sep. 2025, pp. 101–106.
6. M. Domagk, J. Meyer, J. Kilter, R. Maripuu, "Use Cases for Power Quality Data Analysis: Case Study for the Estonian Transmission System," *21st ICHQP*, Chengdu, China, Oct. 2024, pp. 485–490.
7. A. M. Blanco, M. Domagk, J. Meyer, M. Lindner, "Flexible Network Model to Study the Impact of Future Changes in Transmission Systems on Harmonic Levels and Impedance," *CIGRE Session 2024*, Paris, France, Aug. 2024.
8. A. Bracale, P. Caramia, P. De Falco, M. Domagk, J. Meyer, "Probabilistic Forecasting of Current Harmonic Distortions in Distribution Systems," *2023 IEEE PES Innovative Smart Grid Technologies Europe (ISGT Europe)*, Oct. 2023, pp. 1–5.
9. M. Domagk, J. Meyer, K. Scheida, R. Braunstein, E. Traxler, R. Zoll, "Impact of Discontinuous Measurements on the Trend Analysis of Power Quality Parameters," *27th International Conference on Electricity Distribution (CIRED)*, Rome, Italy, Jun. 2023.
10. O. Zyabkina, M. Domagk, E. Strunz, J. Meyer, T. Wang, D. Feng, W. Huang, "Identification of Disturbance Patterns in Long-Term Measurements of Power Quality Characteristics in Chinese Large Cities," *CIRED 2022 Shanghai Workshop*, Sep. 2022, pp. 308–312.
11. M. Domagk, J. Meyer, T. Wang, D. Feng, W. Huang, H. Mayer, S. Wenig, M. Lindner, J.-H. Amrhein, "Trend Analysis for Power Quality Parameters Based on Long-Term Measurement Campaigns," *20th ICHQP*, Naples, Italy, May 2022, pp. 1–6.
12. M. Domagk, I. Y.-H. Gu, J. Meyer, P. Schegner, "Automatic Identification of Different Types of Consumer Configurations by Using Harmonic Current Measurements," *Applied Sciences*, vol. 11, no. 8, 3598, Apr. 2021.

Invited Talks & Presentations (selected)

1. "Challenges of Measurement-Based Determination of Frequency-Dependent Impedance Characteristics," transCampus Project Workshop, King's College London, UK, May 2023.
2. "Analysis and Visualization of Large-Scale Power Quality Monitoring Campaigns," Panel Session on Power Quality Data Analysis, 21st ICHQP, Chengdu, China, Oct. 2024.
3. "Analysis of Large-Scale Power Quality Data Sets – Practical Examples from the Transmission System," OMICRON Diagnostics Conference, Lindau, Germany, Oct. 2025.
4. "Large-Scale Power Quality Measurement Programs – Efficient Implementation, Visualization & Evaluation," ETFS gogreen Seminar, Straubing, Germany, Feb. 2026.

Committee & Organizational Service

Publication Chair, *22nd International Conference on Harmonics and Quality of Power (ICHQP)*, Dresden — 2026

Organizing Committee, *1st ICHQP Workshop 2025*, London, UK — 2025

Awards

Best Student Paper Award, “Trend Identification in Power Quality Measurements,” Australasian Universities Power Engineering Conference (AUPEC), Wollongong, Australia — Sep. 2015